EXPERIMENT : 6

AIM:

Time Series Analysis of Website Traffic.

OBJECTIVE:

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

METHODOLOGY:

1. Load and Clean Data:
    - Import the dataset and parse the Date column as a datetime object.
    - Handle missing values by interpolation.
2. Time Series Decomposition:
    - Decompose the time series data of Page_Views to identify trend, seasonality,and residuals.
3. Visualization:
    - Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate.
    - Use moving averages to smooth the time series and highlight trends.
    - Create seasonal plots to visualize patterns over time.
4. Anomaly Detection:
    - Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
data = pd.read_csv('/content/traffic.csv')

# Combine Date and Time
data['DateTime'] = pd.to_datetime(
    data['Date'].astype(str) + ' ' + data['Time'],
    format='%d %I:%M:%S %p',
    errors='coerce'
)

# Remove invalid dates and sort
data = data.dropna(subset=['DateTime'])
data = data.sort_values('DateTime')
data = data.set_index('DateTime')

# Use Total as the traffic count
data['Total'] = pd.to_numeric(data['Total'], errors='coerce')
data = data.dropna(subset=['Total'])

# Time Series Decomposition
decomposition = seasonal_decompose(
    data['Total'],
    model='additive',
    period=96
)
```

```
decomposition.plot()
plt.show()

# Moving Average
data['Total_MA'] = data['Total'].rolling(window=30).mean()

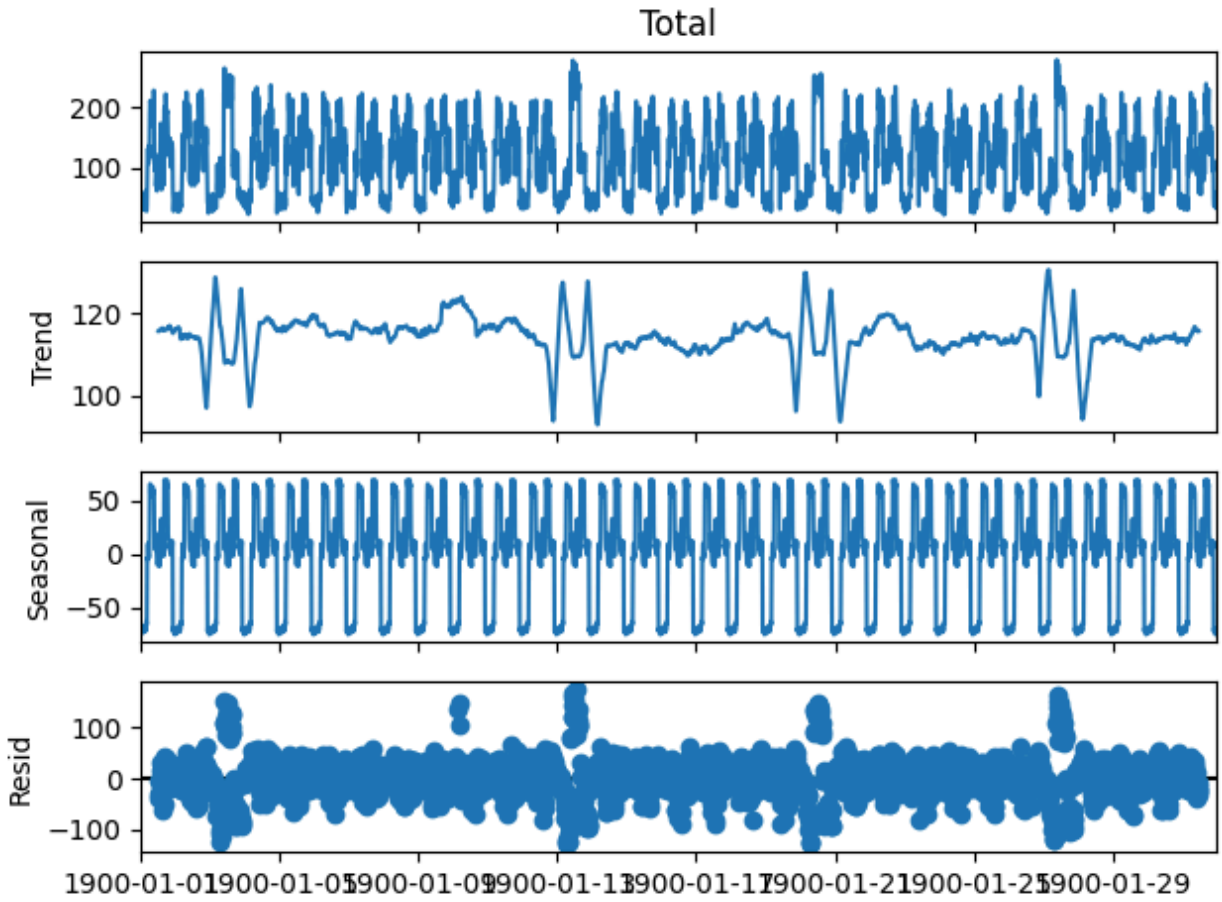
data[['Total', 'Total_MA']].plot(figsize=(12, 5))
plt.title('Traffic Count with Moving Average')
plt.show()

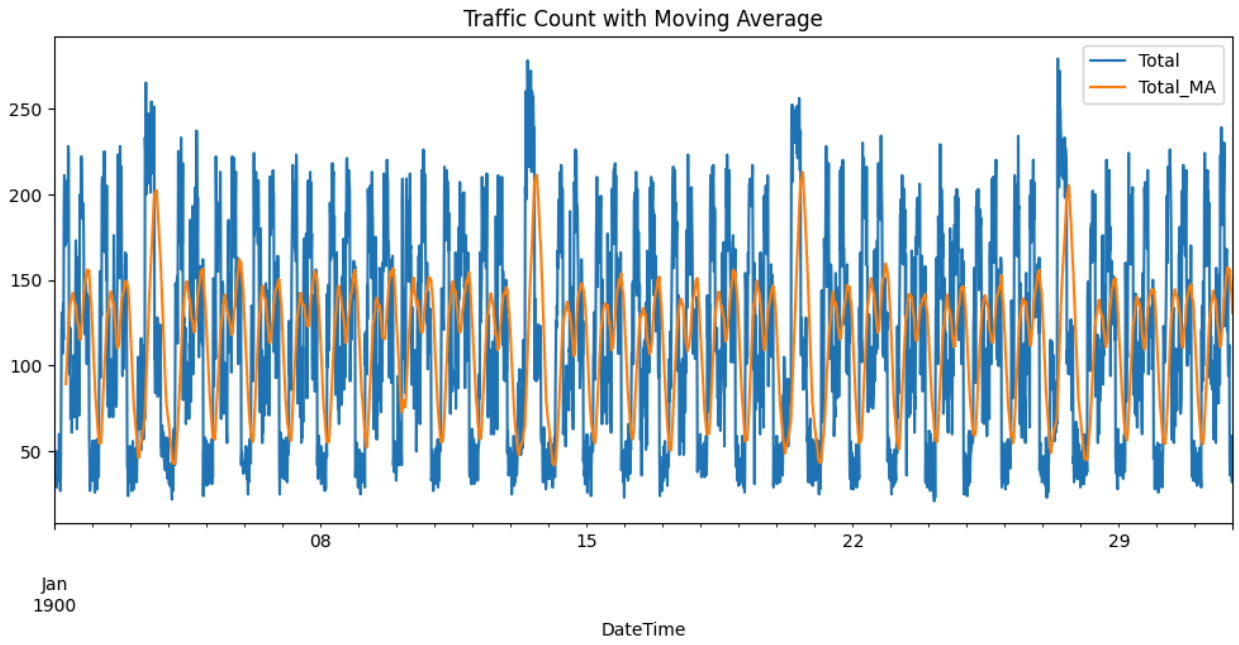
# Anomaly Detection using Z-score
data['Total_Z'] = (
    (data['Total'] - data['Total'].mean())
    / data['Total'].std()
)

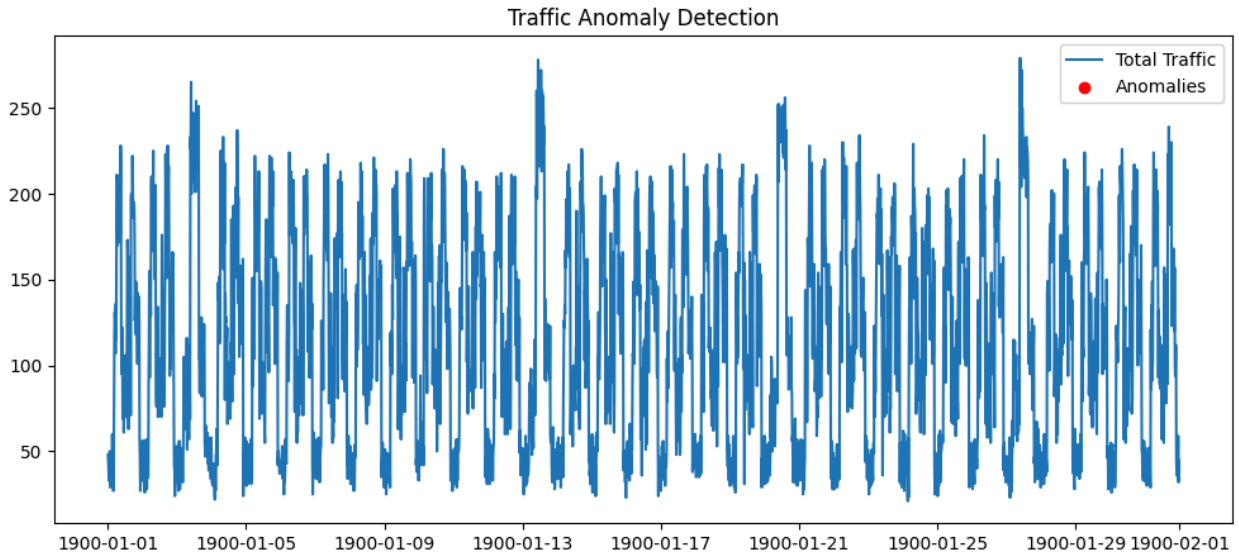
anomalies = data[data['Total_Z'].abs() > 3]

plt.figure(figsize=(12, 5))
plt.plot(data.index, data['Total'], label='Total Traffic')
plt.scatter(
    anomalies.index,
    anomalies['Total'],
    color='red',
    label='Anomalies'
)
plt.title('Traffic Anomaly Detection')
plt.legend()
plt.show()
```

OUTPUT:


Total
200
100
Trend
120
100
Seasonal
50
0
−50
Resid
100
0
−100
1900-01-01
1900-01-05
1900-01-09
1900-01-13
1900-01-17
1900-01-21
1900-01-25
1900-01-29





RESULT:

Time series analysis was successfully performed on the website traffic data to identify trends and seasonal patterns. Anomalies in traffic were also examined to understand unusual changes in website visits over time.